Heat Illness Education in High School Sports and Marching Bands

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SUMMARY
Heat-related illnesses (HRIs) can be life-threatening for high school students who engage in exertional activities. However, targeted HRI education remains limited. This work develops and evaluates a 10-minute, self-paced online educational module structured around six learning objectives derived from published guidance on HRI prevention, recognition, and emergency response. A pilot study tested the module in a randomized controlled trial with 20 high school athletes and marching band members assigned to intervention and control groups. Preliminary results show a positive short-term effect on HRI knowledge and support the feasibility of low-cost online delivery.

KEYWORDS
Heat-related illness, community health education, randomized controlled trial

INTRODUCTION
Heat-related illnesses (HRIs) range from heat cramps and heat exhaustion to potentially fatal heat stroke. They occur when heat exposure, physical exertion, or both exceed the body's ability to regulate temperature. Heat stroke can involve central nervous system dysfunction.

HRI risk is amplified during sports and other vigorous activities in hot and humid conditions. HRIs are among the top three causes of athlete deaths[4]. From 2000 to 2024, at least 77 U.S. athletes died from HRIs, including 65 teenagers (84%)[13]. Over 9,000 high school athletes receive HRI treatment annually[14]. Marching band members face HRI risk in full uniforms while carrying instruments[9].

For both athletes and marching band members, environmental risk factors include high temperature and humidity, direct sun exposure, limited airflow, and radiant heat from sun-heated surfaces (e.g., asphalt and concrete). Other risk factors include insufficient heat acclimatization, dehydration, previous heat illness, poor sleep, intense or prolonged exertion, and heat-trapping uniforms and equipment. Acclimatization is often underestimated; HRIs can occur even before summer in cooler regions[4,9].

Since these factors interact, prevention requires more than hydration alone. It also involves gradual acclimatization, environmental monitoring, adjusted activity intensity and work-to-rest ratios, symptom recognition, and emergency action plans that specify rapid cooling before transport for suspected heat stroke[4,5]. The American Academy of Pediatrics recommends regularly educating students about acclimatization, hydration, honest symptom reporting, recovery, and rest[2].

This work develops a community health intervention focused on HRI prevention, recognition, and emergency response among high school athletes and marching band members. It is delivered through a self-paced online educational module and evaluated through a randomized controlled trial.

BACKGROUND
Heat-related community interventions include education, environmental monitoring, heat warning and communication, hydration and cooling practices, and social support[3,7,12]. Studies suggest that educational interventions can improve heat-related knowledge[3,7], but rigorous evaluations, particularly randomized controlled trials (RCTs), remain limited[3]. A notable exception is Razzak et al. (2022), which evaluates an educational program in low-income communities in Pakistan with an RCT[11]. The present work also uses an RCT but targets high school students engaged in exertional activities.

Student-focused education interventions have often emphasized hydration rather than comprehensive HRI education covering risk factors, symptom recognition, and emergency response[1,8,12]. Taking a broader approach, Li et al. evaluated a program on heatwaves and climate change in two elementary schools in China, with one school assigned to the intervention group and the other to the control group[10]. In contrast, this work evaluates comprehensive HRI education in high school sports and marching bands in the U.S., with students individually randomized to intervention and control groups.

METHODS

Educational Module: This work developed a self-paced online module consisting of a 10-minute, 10-slide narrated presentation and an accompanying handout. The module and assessment are structured around six learning objectives derived from published guidance on HRI prevention, recognition, and emergency response (Table 1)[2,4,5,6]. The module distinguishes students' responsibilities from those of supervising adults and healthcare professionals. Students are not instructed to diagnose HRIs, establish heat safety policies, or independently manage emergencies. Instead, the module emphasizes recognizing danger, stopping activity, reporting concerns, obtaining adult and emergency assistance, and assisting as directed. It also addresses social pressures from teammates and peers that may encourage students to hide symptoms or continue participating.

Table 1 Learning Objectives and Assessment Design

Learning objective (LO)	Operational description
LO1: Identify risk factors	Identify environmental, individual and activity-related factors that increase HRI risk
LO 2: Select prevention strategies	Select appropriate actions before and during activity, including acclimatization, hydration, activity adjustment, rest, and clothing/equipment adjustments
LO 3: Recognize warning signs and stop activity	Recognize common symptoms and observable signs that require the affected student to stop participating and seek help
LO 4: Select reporting and peer-support actions	Report symptoms promptly, resist pressure to continue participating, support symptomatic peers, and remain with them until a supervising adult takes over
LO 5: Recognize possible heat stroke	Recognize altered mental status, loss of coordination, collapse, seizure, and other red flags as signs of a possible medical emergency
LO 6: Select immediate emergency response	Stop activity, alert a supervising adult, activate the emergency plan and 911 response, begin rapid cooling as directed, and avoid unsafe fluid administration

Study Design and Participants: This work conducted an RCT to evaluate the module's effect on HRI knowledge among high school students. Participants in grades 10–12 were recruited from the tennis teams, lacrosse teams, and marching band programs at Bedford High School and Milton High School in Massachusetts. They were randomly assigned to intervention and control groups in a 1:1 ratio through the online study platform. Intervention-group participants completed the assessment immediately after the module, while control-group participants completed the same assessment before receiving access to the module. Information about grade level, activity group, and prior HRI education or experience was collected to characterize the study population. No personal identifiers, such as names, email addresses, student IDs, or IP addresses, were collected. Participation was voluntary.

Assessment: HRI knowledge was measured with an 18-item assessment aligned with the six learning objectives in Table 1. Three items assess each learning objective. Each item is a multiple-choice question. Each correct response receives one point, producing a total score ranging from 0 to 18. All participants were instructed to complete the assessment without consulting the educational handout so that scores reflect unaided knowledge.

Data Analysis: Assessment scores were first summarized with descriptive statistics for the intervention and control groups, including mean and median scores, score range, standard deviation, and IQRs. The intervention effect was estimated using linear regression, with the total assessment score (0–18) as the dependent variable and intervention status as the explanatory variable (0=control, 1=intervention). In this model, the coefficient for intervention status represents the estimated intervention-control mean score difference.

RESULTS

Figures 1 and 2 show the results of an initial pilot with 20 high school athletes and marching band members, with 10 participants in each group. All 20 participants completed the assessment (100% completion rate). The intervention group had a mean score of 17.3 (median: 18, standard deviation: 1.25, standard error: 0.40, 95% confidence interval: 16.41–18.19), compared with 12.2 in the control group (median: 12, SD: 1.75, SE: 0.55, 95% CI: 10.95–13.45). As illustrated in Fig. 1, intervention-group scores are consistently higher on average than control-group scores. Linear regression estimates an intervention effect of 5.1 points on the 18-point assessment, equivalent to 28.3 percentage points (SE: 0.68, 95% CI: 3.67–6.53, $p < 0.001$).

Figure 2 shows higher correct-response rates for the intervention group in all six learning objectives. Intervention-group accuracy ranges from 90% to 100%, compared with 20% to 90% in the control group, with the largest difference for emergency response (LO 6: 93% vs. 20%). Although preliminary, these

patterns suggest that the intervention can improve HRI knowledge across multiple content areas, with a particularly substantial effect on emergency response knowledge.

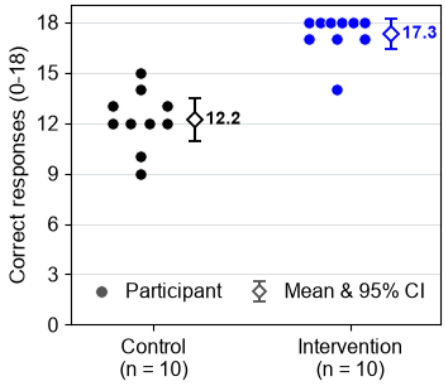


Figure 1 Assessment Scores by Study Group

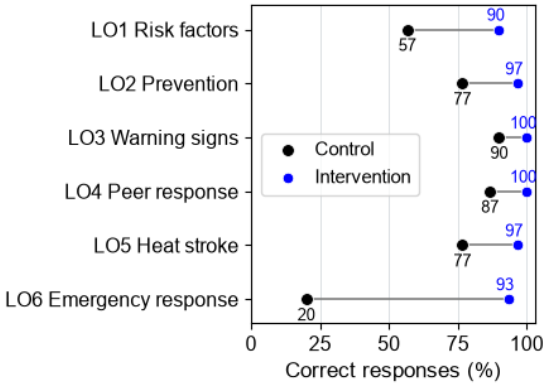


Figure 2 Correct Responses by Learning Objective

CONCLUSION

This work developed an evidence-based, low-cost educational intervention to improve students' ability to prevent, recognize, and respond to HRIs. Preliminary results indicate a substantial short-term effect on HRI knowledge and suggest online implementation feasibility. With support from coaches, athletic and band directors, and other school personnel, the intervention can be incorporated into preseason or early-season heat safety education with minimal additional resources. Given the current small sample, future work should include larger studies across additional schools with follow-up assessments to confirm the findings and examine longer-term knowledge retention.

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